

Vol 9 Chapter 11 — Quantum Spin Liquids

Keeper (author pass — deep math/physics revision)

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Chapter 11 — Quantum Spin Liquids

Level 1 — one sentence

Quantum spin liquids are magnetic systems where geometric frustration prevents conventional magnetic ordering down to $T = 0$, with fractionalized spinon excitations and emergent gauge fields giving exotic topological order — and BST has filed specific order-parameter predictions (Task #107) for the leading candidate materials.

Level 2 — graduate-physicist precision

11.1 Geometric frustration

A simple frustrated system: three spins at triangle vertices with antiferromagnetic Heisenberg coupling. No spin configuration satisfies all three antiferromagnetic bonds simultaneously. Lowest-energy states are degenerate.

Kagome lattice (corner-sharing triangles), pyrochlore lattice (corner-sharing tetrahedra) are quintessentially frustrated. Cause: ground state has macroscopic degeneracy at the classical level $\rightarrow$ quantum effects matter at all temperatures.

11.2 Resonating valence bond (RVB)

Anderson 1973: ground state of frustrated antiferromagnet can be a resonating valence bond state — singlet pairs that resonate among many configurations. No long-range order; gapless or gapped depending on details.

Re-introduced for cuprates by Anderson 1987 in superconductivity context.

11.3 Spinon excitations and fractionalization

In a spin liquid, the elementary excitations are spinons — spin-1/2 quasiparticles that carry only spin and no charge. In contrast to spin-1 magnons of ordered antiferromagnets, spinons can be deconfined.

Spinon deconfinement is a topological-order property: spinons exist in pairs in ordered antiferromagnets (the pair = spin-1 magnon) but separate freely in the spin liquid.

11.4 Emergent gauge fields

Spin liquids typically host emergent gauge fields: - $\mathbb{Z}_2$ gauge field (toric code, gapped spin liquids like kagome candidate) - $U(1)$ gauge field (gapless spin liquids)

Photon-like excitations in $U(1)$ spin liquids = emergent gauge bosons.

11.5 Candidate materials

- Herbertsmithite $\text{ZnCu}_3(\text{OH})_6\text{Cl}_2$ (kagome): no magnetic order down to mK. Leading kagome candidate.
- Pyrochlores $\text{Yb}_2\text{Ti}_2\text{O}_7$, $\text{Tb}_2\text{Ti}_2\text{O}_7$: spin-ice and spin-liquid behaviors

- Triangular Cs_2CuCl_4 : 2D triangular spin liquid candidate
- $\alpha\text{-RuCl}_3$ (Kitaev candidate): leading Kitaev-honeycomb realization

11.6 The Kitaev model

Kitaev 2006: honeycomb lattice with bond-dependent Ising couplings $H = - \sum_{\langle ij \rangle_\gamma} K_\gamma S_i^\gamma S_j^\gamma$. Exactly solvable.

Ground state: $\mathbb{Z}_2$ spin liquid with Majorana fermion + static $\mathbb{Z}_2$ flux. Non-Abelian phase in magnetic field; supports anyons useful for topological quantum computing.

$\alpha\text{-RuCl}_3$ is the leading material realization candidate.

11.7 BST spin liquid predictions (Task #107)

Task #107 (completed): BST has filed substrate-K-type-based predictions for spin liquid order parameters in candidate materials. Specific predictions cataloged in BST notes; full publication awaiting external review.

11.8 K-audit anchors

- Vol 6 Ch 8 (phase transitions, including spin systems)
- Ch 6 of this volume (magnetism)
- Task #107: BST spin liquid predictions

Level 3 — 5th-grader accessibility

Spin liquids are magnetic systems that *refuse* to order even at $T = 0$. The frustration: arrange three spins at triangle vertices, each wanting to anti-align with neighbors — can't be done. Quantum fluctuations prevent any classical order from forming. Excitations are spinons (fractionalized spin-1/2 particles) instead of magnons. Emergent gauge fields arise — the spin liquid looks like a tiny universe with its own photons. Leading candidates: herbertsmithite (kagome lattice), $\alpha\text{-RuCl}_3$ (Kitaev honeycomb). Quantum computing applications: Kitaev's model supports non-Abelian anyons useful for topologically-protected quantum bits.

What comes next

Chapter 12 closes Vol 9.

Where to look this up

- Balents 2010 review *Nature*
- Savary and Balents 2017 review
- BST Task #107